

Lab 1:

1. Hello world.
2. Sum of 3 numbers
- 3.
- 4.
- 5.

Lab 2:

1. Write a program to read a length in inch scale and convert it in the centimeter scale.
2. Write a program to convert a temperature reading in degree Fahrenheit to degree Celsius scale using the formula: $C = (5/9) * (F - 32)$
3. Write a program to read the radius of a circle and calculate its area and circumference.
4. X, Y, Z are the marks of a student. Write a program to find the total and average marks of the student.
5. Write a program to compute the area of the triangle given the values of A, B and C. Area of a triangle is given by the formula $\text{Area} = \sqrt{S(S-A)(S-B)(S-C)}$ where A, B and C are the sides of triangle and $2S = A + B + C$.
6. Evaluate the polynomial:

$$Y = \left(\frac{x-1}{x}\right) + \left(\frac{x-1}{x}\right)^2 / 2 + \left(\frac{x-1}{x}\right)^3 / 3 + \left(\frac{x-1}{x}\right)^4 / 4$$

where the value of x is given from the keyboard.

7. Write a program to calculate the roots of the quadratic equation $ax^2 + bx + c = 0$ where a, b and c are known.

Lab3:

1. Write a program to determine whether a given number is odd or even.
2. Write a program to find out the largest number among four numbers A, B, C and D.
3. Write a program that reads a year from the keyboard and determine whether it is leap year or not.
4. Write a program that computes an employee's gross pay and net pay using the formulas-

$$\text{Gross} = \text{Hours} * \text{Rate}$$

$$\text{Net} = \text{Gross} - \text{Tax}$$

Tax is subtracted from the gross only if an employee earns more than TK.5000. Otherwise deduct no tax. Tax rate is 5% of gross pay.

5. Write a program which receives the score of a student and display the grade according to the following classification:

Grade	Score
A	80 ..100
B	65 .. 79

C	50 .. 64
D	40 .. 49
F	00 .. 39

6. An electricity board charges the following rates to domestic users to discourage the large consumption of energy:

For the first 100 units : Tk. 1.75 per unit

For next 200 units : Tk. 2.25 per unit

Beyond 300 units : Tk. 3.50 per unit

All users are charged a minimum of Tk. 100/=. If the total cost is more than Tk. 1000/= then an additional surcharge of 15% is added.

Write a program to read out the names of users and number of units consumed and print out the charges with names.

7. Write a program that plays the game of “Rock, paper, scissors”. In this game, two players simultaneously say “rock”, “paper” or “scissors”. The winner is one whose choice dominates the other. The rules are: paper dominates (wraps) rock, rock dominates (breaks) scissors, and scissors dominate (cut) paper.

Lab4:

1. Write a program to convert an integer from decimal to binary.
2. Write a program to sort n numbers taken from keyboard.
3. Write a program to calculate row sum and column sum.
4. Write a program that reads a positive integer n and then prints a diamond of asterisks in $2n-1$ rows. For example, if n is 4, then the output would be

```

      *
    * * *
  * * * * *
* * * * * *
  * * * * *
    * * *
      *

```

5. Write a program that reads a positive integer n and then prints a pyramid of numbers in $2n-1$ rows. For example, if n is 4, then the output would be

```

      1
    1 2 1
  1 2 3 2 1

```

6. Binary/Linear search.

Lab 5:

1. Convert the characters of a string into opposite case.
2. Calculate the frequency of characters of a string.
3. Palindrom
4. Anagram
5. Calculate the sum of 2 big integers (string).
6. String sort

Lab 6:

1. Method overloading
2. Single Inheritance
3. Multilevel Inheritance
4. Hierarchical Inheritance
5. Contractor
6. Final Keyword
7. Super Keyword
8. Static Keyword
9. Abstract Keyword
10. Method overriding

Lab 7: Exception and Thread

1. Try-catch-finally
2. Create a new exception
3. Thread using Thread class
4. Thread using Runnable interface
5. Thread sleep method
6. Thread priority
7. Suspend-resume

Lab 8: Applet

1. Display 'Hello World' in an applet.
2. Display a circle, triangle and rectangle.
3. Draw smiling face.
4. Draw Bangladeshi flag.
5. Draw 'A', 'B'.
6. Draw Shahid minar
7. Draw Sritishoudo.
8. Draw scrolling text

Lab 9: AWT- event –layout manager

1. Design a GUI frame with a text field and a button.
2. Design a small/simple calculator.
3. Analog clock
4. Digital clock
5. Tic-tack-toe

Lab 10: Input output

1. Create a file and write name, ID and dept. in the file.
2. Display from a data file

Lab 11: Java network and Database

1. Server-client chat
2. Who is server
3. InetAddress
4. Data send using datagram
5. Database program to display data from a table
6. Database program to insert/update data from a table
7. Delete data from a table
8. Search a data from a table